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UNIVERSITY OF CAMBRIDGE | Cavendish Laboratory
Department of Physics

Mathematical Methods for Scientific Computing

Multivariable Calculus

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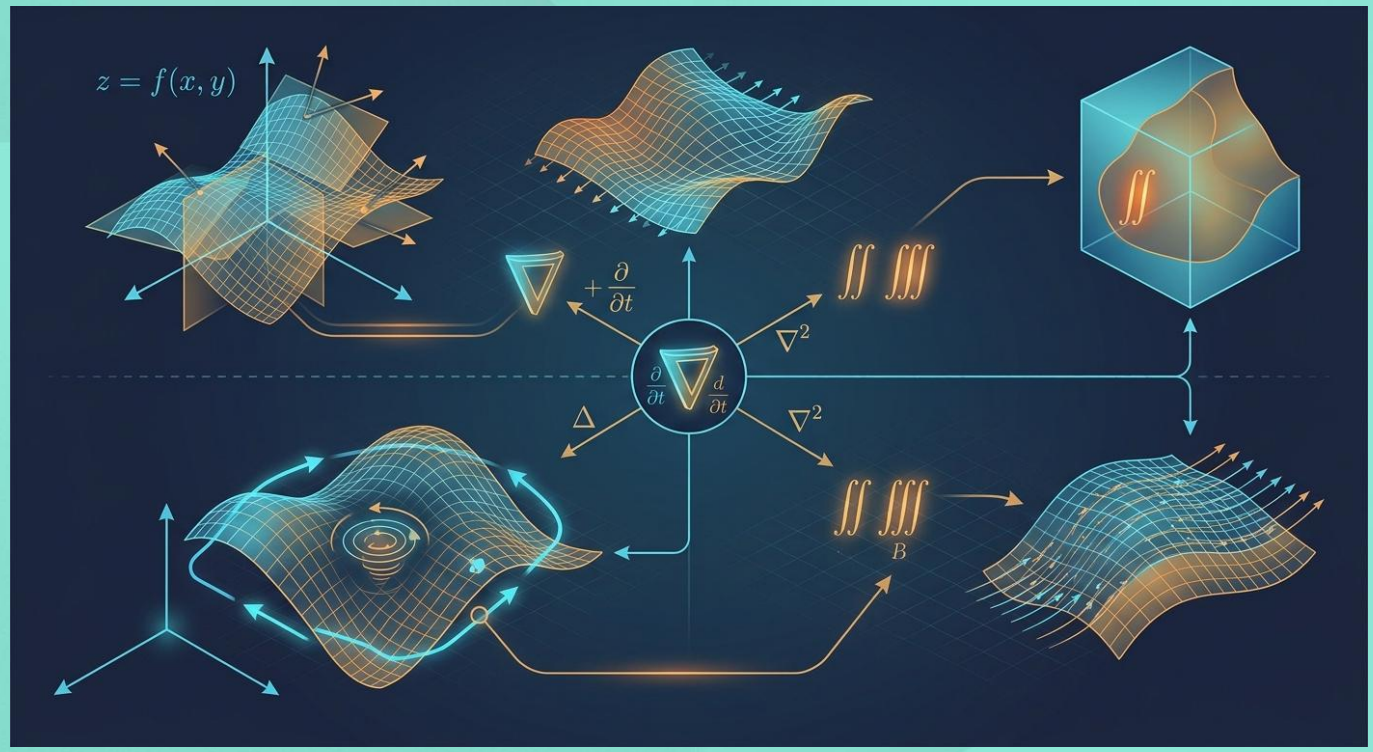
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Outline

- 1. Multivariable calculus
- 2. Thermodynamics and partial derivatives
- 3. Derivatives and integrals

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Multivariable calculus



Coupled systems of equations

- Scientific computing is a valuable tool for complex physical problems
- Such problems rarely have a single, well-behaved variable, but instead we must consider complex, coupled multivariable systems
- And coincidentally, my favourite equations, the compressible Euler equations are just such a system

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$

- These are coupled equations – we cannot split them into three individual equations
- They also have rather more than three variables it seems

Independent and dependent variables

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$

- For this to be a closed system of equations, we need to have three **independent variables**
- A good choice here is density, ρ, momentum, $\rho\mathbf{v}$, and total energy, E – all of which are known to be quantities which are conserved
- For a closed system of equations, we also need to define velocity, $\mathbf{v}$, and pressure, p, in terms of the independent variables – and these are **dependent variables**

Coupled systems of equations

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\mathbf{v} = \frac{\rho \mathbf{v}}{\rho}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

$$p = p(\rho, \varepsilon)$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{v}] = 0$$

$$E = \rho \varepsilon + \tfrac{1}{2} \rho v_i v_i$$

- The choice of independent variables is not unique – as long as the system is **closed** by three variables, we can label these as the independent variables
- For some applications, the actual choice of independent variables doesn't really matter – we just need to know the equations can be solved
- However, if we want to start manipulating our equations, and performing additional calculus, we need to be certain we are consistent with how variables are treated

Closure and derivatives

$$\mathbf{v} = \frac{\rho \mathbf{v}}{\rho}, \qquad p = p(\rho, \varepsilon), \qquad E = \rho\varepsilon + \frac{1}{2}\rho v_i v_i$$

- When we need to manipulate these equations, we frequently find ourselves taking derivatives with respect to other variables
- Before doing this, we need to be sure which independent variables we have
- For example, in the situation with ρ, $\rho\mathbf{v}$ and E being the independent variables,

$$\frac{\partial E}{\partial(\rho v_x)} = \left.\frac{\partial E}{\partial(\rho v_x)}\right|_{\rho,\rho v_y,\rho v_z=\text{const}} = 0$$

- Another parameterisation might use ρ, ε and ρv as independent variables

$$\frac{\partial E}{\partial(\rho v_x)} = \left.\frac{\partial E}{\partial(\rho v_x)}\right|_{\rho,\varepsilon,\rho v_y,\rho v_z=\text{const}} = v_x$$

Outside of these equations, you could consider taking the derivative of a function $f(x, y, z)$ with respect to x, compared with the derivative of this function on the surface of a cone, $f\left(x, y, z = \sqrt{x^2 + y^2}\right)$ - in one of these, z is treated as constant, in the other it is not

Switching between multiple variables

- When it comes to physical systems, there are a large number of variables which can be used to describe properties, e.g. pressure, temperature, volume, density, number density, entropy, energy, momentum, velocity, enthalpy and heat (not an exhaustive list)
- These are not all independent variables, and often we will be able to write one in terms of one or more of the other variables
- For example, the ideal gas law can be expressed as

$$pV = nRT \quad \text{or} \quad p = (\gamma - 1)\rho\varepsilon$$

- More generally, we want equations which don't assume a form for these these relationships, in which case, we know that we could write

$$p = p(V, T) \quad \text{or} \quad p = p(\rho, \varepsilon)$$

- When analysing properties of a material, we may find we need to take derivatives with respect to variables we did not use in our formulation, and for this we need the **multivariable chain rule**

Speed of sound

- An example, which motivates why we might need to worry about multivariable calculus, is the speed of sound, which is defined by

$$c_s^2 = \left.\frac{\partial p}{\partial \rho}\right|_s$$

- This expression would be fine if we choose our independent variables to be p, ρ and s (pressure, density and entropy)
- We'd also need to know the function $p = p(\rho, s)$
- Entropy is not a nice quantity to work with, it can't be directly measured, which makes identifying a relationship $p = p(\rho, s)$ difficult; it is far more common to find $p = p(V, T) \quad \text{or} \quad p = p(\rho, \varepsilon)$
- Because density and entropy are independent variables, that meas we could write the other variables in terms of ρ and s, i.e. we know we have $p = p(V(\rho, s), T(\rho, s)) \quad \text{or} \quad p = p(\rho, \varepsilon(\rho, s))$
- Technically, we actually have here $p = p(V(\rho), T(\rho, s))$ since $V = M/\rho$

Multivariable chain rule

- If we have $p = p(V(\rho, s), T(\rho, s))$, and assume we don't know that V is not actually a function of s

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \left.\frac{\partial V}{\partial \rho}\right|_s + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s, \quad \left.\frac{\partial p}{\partial s}\right|_\rho = \left.\frac{\partial p}{\partial V}\right|_T \left.\frac{\partial V}{\partial s}\right|_\rho + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial s}\right|_\rho$$

Multivariable chain rule

- If we have $p = p(V(\rho, s), T(\rho, s))$, and assume we don't know that V is not actually a function of s

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \left.\frac{\partial V}{\partial \rho}\right|_s + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s, \quad \left.\frac{\partial p}{\partial s}\right|_\rho = \left.\frac{\partial p}{\partial V}\right|_T \left.\frac{\partial V}{\partial s}\right|_\rho + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial s}\right|_\rho$$

- If we do actually have $p = p(V(\rho), T(\rho, s))$

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s, \quad \left.\frac{\partial p}{\partial s}\right|_\rho = \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s$$

- If we have $p = p(\rho, \varepsilon(\rho, s))$

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial \rho}\right|_\varepsilon + \left.\frac{\partial p}{\partial \varepsilon}\right|_\rho \left.\frac{\partial \varepsilon}{\partial \rho}\right|_s, \quad \left.\frac{\partial p}{\partial s}\right|_\rho = \left.\frac{\partial p}{\partial \varepsilon}\right|_\rho \left.\frac{\partial \varepsilon}{\partial s}\right|_\rho$$

Reciprocal and cyclic rules

- We are about to see that we may need to manipulate the partial derivative expressions to make them easier to compute
- To help, we can derive useful relationships

1. The reciprocal rule

$$\left.\frac{\partial z}{\partial y}\right|_x = \frac{1}{\left.\frac{\partial y}{\partial z}\right|_x}$$

2. The cyclic rule

$$\left.\frac{\partial z}{\partial x}\right|_y \left.\frac{\partial x}{\partial y}\right|_z \left.\frac{\partial y}{\partial z}\right|_x = -1$$

Higher derivatives

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s$$

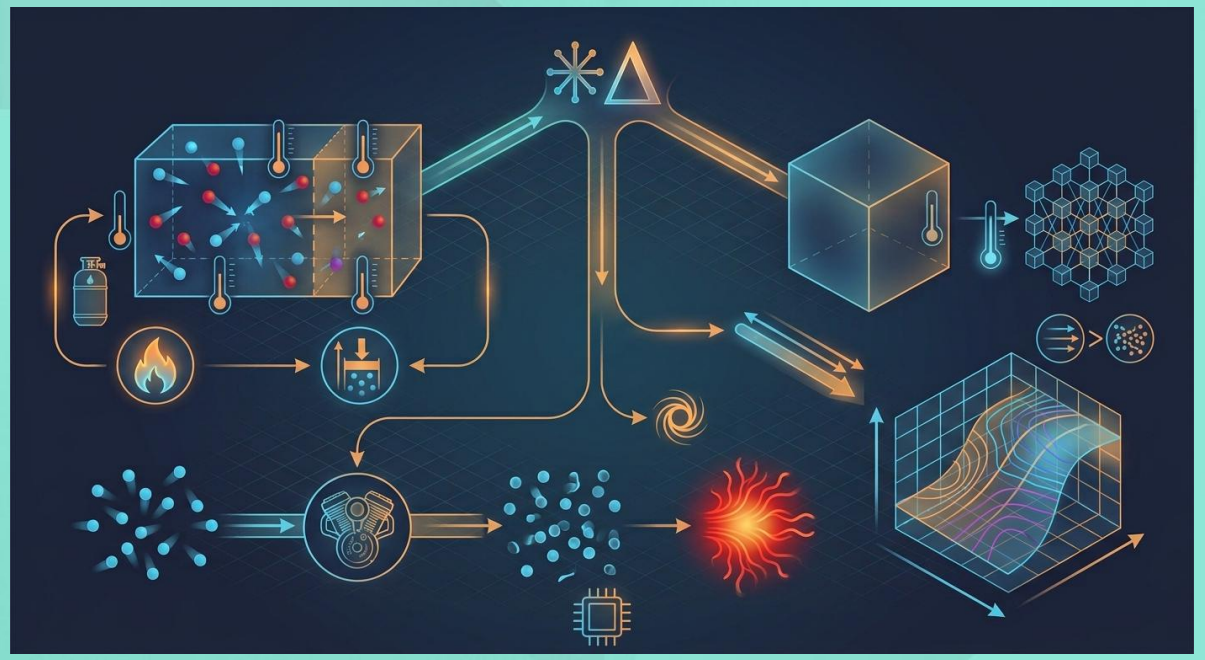
- If we want the second derivative, e.g. $\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s$, we could derive a general expression

$$\begin{aligned}\left.\frac{\partial^2 p}{\partial \rho^2}\right|_s = &\left[\frac{\partial}{\partial V}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \frac{\partial}{\partial T}\left(\left.\frac{\partial p}{\partial V}\right|_T\right)\Bigg|_V \left.\frac{\partial T}{\partial \rho}\right|_s\Bigg] \frac{\mathrm{d}V}{\mathrm{d}\rho} \\ &+ \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}^2 V}{\mathrm{d}\rho^2} \\ &+ \left[\frac{\partial}{\partial V}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \frac{\partial}{\partial T}\left(\left.\frac{\partial p}{\partial T}\right|_V\right)\Bigg|_V \left.\frac{\partial T}{\partial \rho}\right|_s\Bigg] \left.\frac{\partial T}{\partial \rho}\right|_s \\ &+ \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial^2 T}{\partial \rho^2}\right|_s\end{aligned}$$

- In practice, this is cumbersome, and prone to typos – instead, it is usually better to compute the first derivative, write it in terms of sensible variables (e.g. ρ and s), and take a much simpler second derivative

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Thermodynamics and partial derivatives



Another look at the speed of sound

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s$$

- In order to compute the speed of sound, we need to compute various terms here
- Some are straightforward; for example, we know $V = 1/\rho$
- Some are material properties, which, when we apply this to our choice of material, we would expect to know, e.g. pressure may obey thethe ideal gas law

$$p = \frac{nRT}{V}$$

- This is an **equation of state**, which, depending on your stream, you may well see covered in more detail
- Within this expression, however, are quantities we don't want to work with if we can help it; for many materials we may not know the function $T = T(\rho, s)$, and therefore cannot compute $\left.\frac{\partial T}{\partial \rho}\right|_s$

Thermodynamics and closure

- Throughout the slides in this topic, we have been introducing new variables, despite our motivation being closure of our systems of equations
- For those counting, we now have ρ, $\rho\boldsymbol{v}$, E, ε, p, $\mathbf{v}$, T, V and s
- We've also introduced three conservation laws, an equation of state, and have defined ε, $\mathbf{v}$ and V in terms of other variables
- We then have 7 relationships for 9 variables, this is not a closed system
- The remaining closure then comes from thermodynamics (with a little bit of experimental measurement)
- The **first law of thermodynamics** can be written as,

$$\mathrm{d}Q = \mathrm{d}U + p\,\mathrm{d}V \qquad \text{or} \qquad \mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$$

Thermodynamic quantities

$$\mathrm{d}Q = \mathrm{d}U + p\,\mathrm{d}V \qquad \mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$$

- The laws of thermodynamics may seem to introduce many new variables, rather than close the system
- However, two of these variables, U and S are directly related to ε and s
- ε and s are internal energy and entropy **per unit mass**
- U and S are **total** internal energy and entropy, and for a given material; we simply divide by, or multiply by, mass (a constant) to convert between these pairs of variables
- We have also introduced heat, Q, which is another quantity that is not natural to work with, but at least it came with an additional equation too
- We still need one more closure equation though, and it may not be immediately clear how we use these relationships, which relate changes of quantities, rather than the quantities themselves

Maxwell's relations

$$\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$$

- This expression suggests that internal energy is a function of volume and entropy, $U = U(V, S)$
- This means that the multivariable chain rule applies

$$\mathrm{d}U = \frac{\partial U}{\partial S}\Big|_V \mathrm{d}S + \frac{\partial U}{\partial V}\Big|_S \mathrm{d}V$$

- Then, by direct comparison

$$\frac{\partial U}{\partial S}\Big|_V = T, \qquad \frac{\partial U}{\partial V}\Big|_S = -p$$

Maxwell's relations

$$\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$$

- We can then take additional derivatives of these terms

$$\frac{\partial^2 U}{\partial V \partial S} = \left.\frac{\partial T}{\partial V}\right|_S, \qquad \frac{\partial^2 U}{\partial S \partial V} = -\left.\frac{\partial p}{\partial S}\right|_V$$

- This is one of Maxwell's relations, others can be derived from:
  - **Helmholtz free energy**

$$F = U - TS \implies \mathrm{d}F = -S\,\mathrm{d}T - p\,\mathrm{d}V$$

  - **Enthalpy**

$$H = U + pV \implies \mathrm{d}H = T\,\mathrm{d}S + V\,\mathrm{d}p$$

  - **Gibbs free energy**

$$G = H - TS \implies \mathrm{d}G = -S\,\mathrm{d}T + V\,\mathrm{d}p$$

Maxwell's relations

- By following the steps of matching chain rule quantities with each of these relationships, and then taking a second derivative, we obtain all of Maxwell's relations

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial p}{\partial S}\right|_V \qquad \left.\frac{\partial S}{\partial V}\right|_T = \left.\frac{\partial p}{\partial T}\right|_V$$

$$\left.\frac{\partial T}{\partial p}\right|_S = \left.\frac{\partial V}{\partial S}\right|_p \qquad \left.\frac{\partial S}{\partial p}\right|_T = -\left.\frac{\partial V}{\partial T}\right|_p$$

- These relationships allow us to rewrite terms where we don't like the variable we are dealing with
- They are particularly useful for avoiding entropy, either as the variable we are keeping constant, or as one of the variables in a partial derivative
- In order to fully close our system, we still need one more relationship

Specific heat capacity

- Specific heat capacity is a measure of the amount of heat required to raise the temperature of a material (measured per unit mass)
- This is dependent on how the heat is added, two typical specific heats are considered
- **Specific heat at constant volume**, c_V, considers the amount of heat required if the volume of material does not change whilst temperature rises (but the pressure, or stress, within the material does)
- **Specific heat at constant pressure**, c_p, considers the amount of heat required if the pressure (or stress) does not change, but instead temperature rise causes volume to change

Specific heat capacity

$$c_V = \frac{1}{M}\left.\frac{\mathrm{d}Q}{\mathrm{d}T}\right|_V, \qquad c_p = \frac{1}{M}\left.\frac{\mathrm{d}Q}{\mathrm{d}T}\right|_p$$

- Typically, actual specific heats are calculated either from first-principles atomistic calculation, or measured experimentally
- Either way, we can assume they are a known function of two independent thermodynamic variables for any given material
- But if we know c_V, we can show

$$U = M\int c_V\,\mathrm{d}T \quad \text{or} \quad U = Mc_VT$$

- The second expression requires that c_V is a constant, which for some materials, under conditions applicable to a simulation, is a valid assumption
- We can relate energy to temperature without worrying about heat (or, technically, hiding it in a known material parameter, c_V)

Volumetric quantities

$$U = Mc_V T$$

- Many of the laws of thermodynamics we have considered work with mass, M, and volume, V
- When dealing with continuum systems, typically we work with quantities per unit volume, e.g. density measured in kg/m³, or quantities per unit mass, e.g. specific internal energy measured in J/kg
- For example, we have (total) internal energy, U, internal energy (per unit volume), $e = U/V$, or specific internal energy, $\varepsilon = U/(\rho V)$
- Note of caution – both U and e are frequently referred to as 'internal energy' in the literature!
- It is common to then see the equivalent relationship between specific internal energy and temperature,

$$\varepsilon = c_V T$$

Dimensional analysis

- Different quantities can get confusing, especially with things like total entropy, S, or entropy per unit mass, s
- The literature cannot be guaranteed to be consistent
- When in doubt, dimensional analysis can help a lot
- The units of total internal energy are Joules,

$$[U] = \mathrm{J} = \mathrm{kg\,m^2\,s^{-2}}$$

- We could be more formal, and take units out of the expression too

$$[U] = \mathrm{M\,L^2\,T^{-2}}$$

- We can confirm that this is the quantity given by $U = Mc_V T$, which either have obvious units (M, T) or are easy to look up (c_V)

Back to the speed of sound (again)

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \left.\frac{\partial p}{\partial V}\right|_T \frac{\mathrm{d}V}{\mathrm{d}\rho} + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial \rho}\right|_s$$

- Now we have thermodynamics to simplify things, we can return to our speed of sound example
- Since all our thermodynamic rules were in terms of V and not ρ, and we know how they are related, we can write

$$c_s^2 = -\frac{1}{\rho^2}\left(\left.\frac{\partial p}{\partial V}\right|_T + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial V}\right|_S\right)$$

- Most of this expression is now easy to evaluate with an equation of state, $p = p(V, T)$, but the last term needs work

Back to the speed of sound (again)

$$c_s^2 = -\frac{1}{\rho^2}\left(\left.\frac{\partial p}{\partial V}\right|_T + \left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial V}\right|_S\right)$$

- Using first the cyclical and reciprocal rules, we get

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial S}{\partial V}\right|_T \left.\frac{\partial T}{\partial S}\right|_V$$

- Then, one of Maxwell's relations, and some simplification gets

$$\left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial S}\right|_V = -\left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial U}\right|_V \left.\frac{\partial U}{\partial S}\right|_V = -T\left.\frac{\partial p}{\partial T}\right|_V \left.\frac{\partial T}{\partial U}\right|_V$$

- And since we saw how, if we know the specific heat at constant volume, we can find $U = U(V,T)$, we can now hopefully compute all derivatives in this expression

Example for the speed of sound

- The eventual expression for the speed of sound reads

$$c_s^2 = -\frac{1}{\rho^2}\left[\left.\frac{\partial p}{\partial V}\right|_T - T\left(\left.\frac{\partial p}{\partial T}\right|_V\right)^2 \left.\frac{\partial T}{\partial U}\right|_V\right]$$

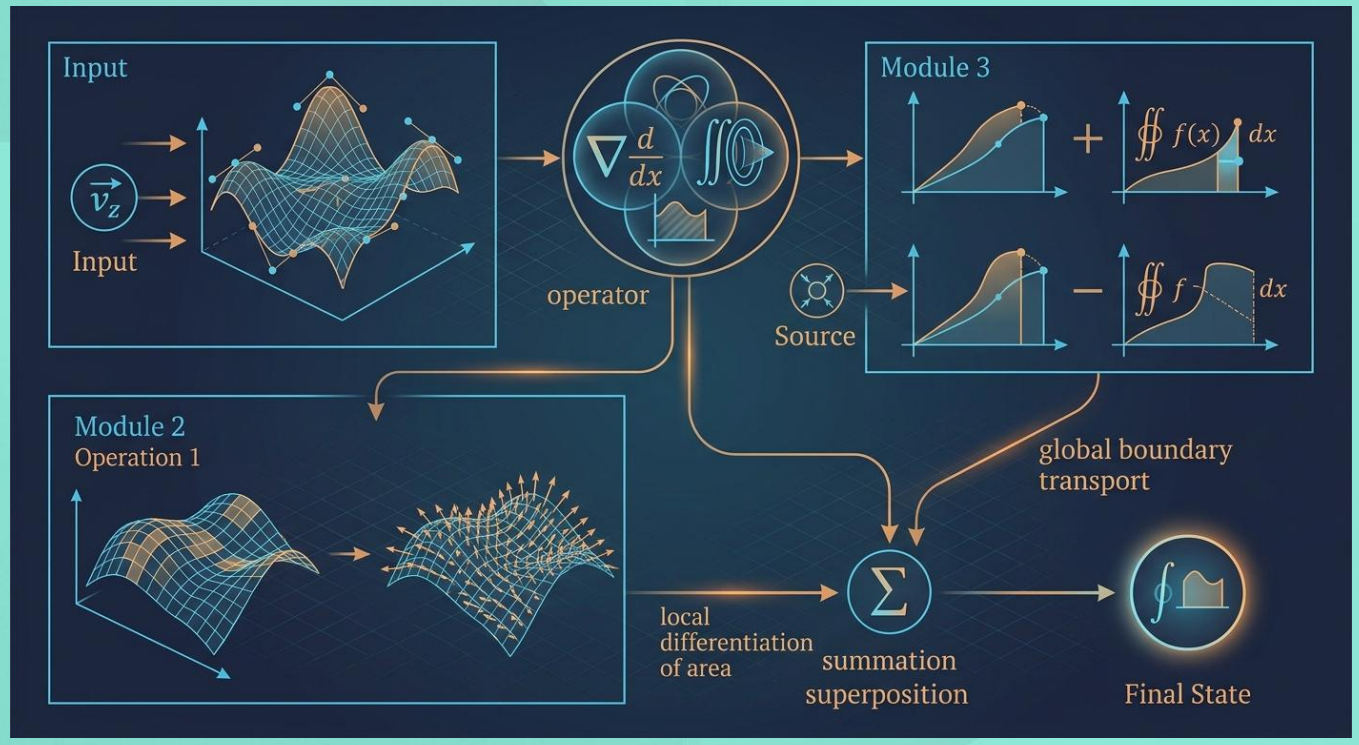
- If we use the ideal gas law (which has constant c_V), we have

$$p = \frac{nRT}{V}, \qquad T = \frac{U}{Mc_V}$$

- This gives a speed of sound

$$c_s^2 = \frac{p}{\rho}\left(1 + \frac{nR}{Mc_V}\right)$$

Derivatives and integrals



Derivatives and integrals

- An interesting application of multivariable calculus is when you want to take the integral of a function with respect to one variable, and a derivative with respect to the other
- For example, consider

$$G(t) = \int_a^b f(t, x)\,\mathrm{d}x$$

- This could represent the total energy at time t within our domain $x \in [a, b]$, and it could be useful to compute $\frac{\mathrm{d}G}{\mathrm{d}t}$
- In this case, this would be trivial, compute the integral first, and then the derivative. But what if the domain was not fixed, and we had

$$G(t) = \int_{a(t)}^{b(t)} f(t, x)\,\mathrm{d}x$$

Why might we have this?

$$G(t) = \int_{a(t)}^{b(t)} f(t, x)\,\mathrm{d}x$$

- One possible application of this is a piston; fuel is compressed by the piston, hence the limits of the volume inside the piston change
- Additionally, although it is a (hopefully) closed domain, the piston exerts pressure, and this can cause fuel to ignite, hence energy is not constant in this domain
- In order to understand the behaviour of this piston, it is reasonable to assume we might be interested in $\frac{\mathrm{d}G}{\mathrm{d}t}$, where G is the total energy within the fuel, and we want to know how this changes (note – energy within the fuel is not conserved, because the combustion of fuel generates kinetic energy in the piston, which removes it from the domain)
- How would we compute this?

Leibniz's integral rule

$$G(t) = \int_{a(t)}^{b(t)} f(t, x)\,\mathrm{d}x$$

- If we're lucky, we can compute the integral first and then take the derivative – but a more general approach exists
- Leibniz's integral rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{a(t)}^{b(t)} f(t, x)\,\mathrm{d}x\right) = f(t, b(t))\frac{\mathrm{d}b}{\mathrm{d}t} - f(t, a(t))\frac{\mathrm{d}a}{\mathrm{d}t} + \int_{a(t)}^{b(t)} \frac{\partial f(t,x)}{\partial t}\,\mathrm{d}x$$

- This lets us reverse the order of operations – differentiating first, then integrating where necessary
- Sometimes this will be easier to compute, and a direct application (relevant to the continuum stream) is behaviour across a shock wave